Finite Difference Schemes for the Black–Scholes Equation

Explicit, Implicit and Crank–Nicolson methods using the course-index notation

1 Starting point: the forward-time Black–Scholes problem

Following the derivation of the Black–Scholes equation, the option value $V = V(S,t)$ satisfies

$$\boxed{\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.} \tag{1}$$

This equation is naturally written as a *terminal value problem*, because the payoff is known at maturity:

$$V(S,T) = V_T(S). \tag{2}$$

To apply finite difference methods forward in time, introduce the same change of time direction used in the notes:

$$\boxed{U(S,t) := V(S,T-t).} \tag{3}$$

Then

$$\frac{\partial U}{\partial t}(S,t) = -\frac{\partial V}{\partial t}(S,T-t), \qquad \frac{\partial U}{\partial S}(S,t) = \frac{\partial V}{\partial S}(S,T-t), \qquad \frac{\partial^2 U}{\partial S^2}(S,t) = \frac{\partial^2 V}{\partial S^2}(S,T-t). \tag{4}$$

Therefore, from (1),

$$-\frac{\partial V}{\partial t} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV, \tag{5}$$

and so the forward-time form is

$$\boxed{\frac{\partial U}{\partial t} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + rS\frac{\partial U}{\partial S} - rU.} \tag{6}$$

On the truncated computational domain

$$R_V^T = \{(S,t) : 0 < S < S^*,\ 0 \le t \le T\}, \tag{7}$$

we solve

$$\boxed{\begin{cases} \dfrac{\partial U}{\partial t} = \dfrac{\sigma^2}{2} S^2 \dfrac{\partial^2 U}{\partial S^2} + rS\dfrac{\partial U}{\partial S} - rU, & (S,t) \in R_V^T, \\ U(S,0) = V(S,T), & S \in [0,S^*], \\ U(0,t) = V(0,T-t), & t \in [0,T], \\ U(S^*,t) = V(S^*,T-t), & t \in [0,T]. \end{cases}} \tag{8}$$

Equivalently, using the notation

$$U(S,0) = u_0(S), \qquad U(0,t) = u_a(t), \qquad U(S^*,t) = u_b(t), \tag{9}$$

we have the initial-boundary value problem

$$\boxed{\begin{cases} \dfrac{\partial U}{\partial t} = \dfrac{\sigma^2}{2} S^2 \dfrac{\partial^2 U}{\partial S^2} + rS\dfrac{\partial U}{\partial S} - rU, \\ U(S,0) = u_0(S), \\ U(0,t) = u_a(t), \\ U(S^*,t) = u_b(t). \end{cases}} \tag{10}$$

2 Grid notation from the course notes

We now discretise the rectangle $[0, S^*] \times [0, T]$ using the index notation of the course notes:

$$\boxed{S_i = h_S i, \qquad i = 0, 1, \dots, N_S, \qquad h_S = \frac{S^*}{N_S}.} \tag{11}$$

For time,

$$\boxed{t_j = h_t j, \qquad j = 0, 1, \dots, N_t, \qquad h_t = \frac{T}{N_t}.} \tag{12}$$

The numerical approximation is denoted by

$$\boxed{U_{i,j} \approx U(S_i, t_j).} \tag{13}$$

The boundary values are known:

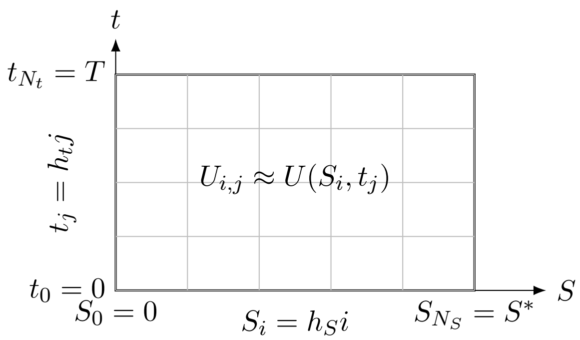
$$U_{0,j} = u_a(t_j), \qquad U_{N_S,j} = u_b(t_j), \qquad j = 0, 1, \dots, N_t. \tag{14}$$

The initial values are also known:

$$U_{i,0} = u_0(S_i), \qquad i = 0, 1, \dots, N_S. \tag{15}$$

The unknown interior values at time level j are

$$U_{1,j}, U_{2,j}, \dots, U_{N_S-1,j}. \tag{16}$$



3 Finite difference approximations from Taylor expansions

We now derive the finite difference formulas directly from Taylor expansions, starting from $U(S, t)$. When expanding in S, the time variable t is fixed.

3.1 Forward difference for $\partial U/\partial S$

Expanding $U(S+h_S,t)$ around (S,t),

$$U(S+h_S,t) = U(S,t) + h_S\frac{\partial U}{\partial S}(S,t) + \frac{h_S^2}{2}\frac{\partial^2 U}{\partial S^2}(S,t) + \frac{h_S^3}{6}\frac{\partial^3 U}{\partial S^3}(S,t) + O(h_S^4). \tag{17}$$

Therefore,

$$\frac{U(S+h_S,t) - U(S,t)}{h_S} = \frac{\partial U}{\partial S}(S,t) + \frac{h_S}{2}\frac{\partial^2 U}{\partial S^2}(S,t) + O(h_S^2), \tag{18}$$

and hence

$$\boxed{\frac{\partial U}{\partial S}(S_i,t_j) = \frac{U_{i+1,j} - U_{i,j}}{h_S} + O(h_S).} \tag{19}$$

3.2 Backward difference for $\partial U/\partial S$

Similarly,

$$U(S-h_S,t) = U(S,t) - h_S\frac{\partial U}{\partial S}(S,t) + \frac{h_S^2}{2}\frac{\partial^2 U}{\partial S^2}(S,t) - \frac{h_S^3}{6}\frac{\partial^3 U}{\partial S^3}(S,t) + O(h_S^4). \tag{20}$$

Thus,

$$\boxed{\frac{\partial U}{\partial S}(S_i,t_j) = \frac{U_{i,j} - U_{i-1,j}}{h_S} + O(h_S).} \tag{21}$$

3.3 Centered difference for $\partial U/\partial S$

Subtracting the two Taylor expansions,

$$U(S+h_S,t) - U(S-h_S,t) = 2h_S\frac{\partial U}{\partial S}(S,t) + \frac{h_S^3}{3}\frac{\partial^3 U}{\partial S^3}(S,t) + O(h_S^5). \tag{22}$$

Therefore,

$$\boxed{\frac{\partial U}{\partial S}(S_i,t_j) = \frac{U_{i+1,j} - U_{i-1,j}}{2h_S} + O(h_S^2).} \tag{23}$$

3.4 Centered difference for $\partial^2 U/\partial S^2$

Adding the two Taylor expansions,

$$U(S+h_S,t) + U(S-h_S,t) = 2U(S,t) + h_S^2\frac{\partial^2 U}{\partial S^2}(S,t) + \frac{h_S^4}{12}\frac{\partial^4 U}{\partial S^4}(S,t) + O(h_S^6). \tag{24}$$

Therefore,

$$\boxed{\frac{\partial^2 U}{\partial S^2}(S_i,t_j) = \frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{h_S^2} + O(h_S^2).} \tag{25}$$

3.5 Forward difference for $\partial U/\partial t$

Expanding in time, with S fixed,

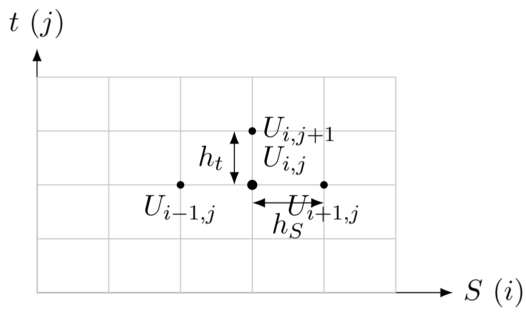
$$U(S, t + h_t) = U(S, t) + h_t \frac{\partial U}{\partial t}(S, t) + \frac{h_t^2}{2} \frac{\partial^2 U}{\partial t^2}(S, t) + O(h_t^3). \tag{26}$$

Hence,

$$\boxed{\frac{\partial U}{\partial t}(S_i, t_j) = \frac{U_{i,j+1} - U_{i,j}}{h_t} + O(h_t).} \tag{27}$$

Finally, since $S_i = h_S i$, we will use

$$\boxed{\frac{S_i}{h_S} = i, \qquad \frac{S_i^2}{h_S^2} = i^2.} \tag{28}$$



4 Explicit method

Starting from

$$\frac{\partial U}{\partial t} = \frac{\sigma^2}{2} S^2 \frac{\partial^2 U}{\partial S^2} + rS \frac{\partial U}{\partial S} - rU, \tag{29}$$

we evaluate all spatial finite differences at time level j:

$$\frac{U_{i,j+1} - U_{i,j}}{h_t} = \frac{\sigma^2}{2} S_i^2 \left(\frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{h_S^2} \right) + rS_i \left(\frac{U_{i+1,j} - U_{i-1,j}}{2h_S} \right) - rU_{i,j}. \tag{30}$$

Multiplying by h_t and isolating $U_{i,j+1}$,

$$U_{i,j+1} = U_{i,j} + \frac{\sigma^2}{2} S_i^2 h_t \left(\frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{h_S^2} \right) \tag{31}$$

$$+ rS_i h_t \left(\frac{U_{i+1,j} - U_{i-1,j}}{2h_S} \right) - rh_t U_{i,j}. \tag{32}$$

Collecting terms and using $S_i/h_S = i$,

$$\boxed{\begin{aligned} U_{i,j+1} = {} & \frac{h_t}{2} \big(\sigma^2 i^2 - ri\big) U_{i-1,j} \\ & + \big(1 - \sigma^2 i^2 h_t - rh_t\big) U_{i,j} \\ & + \frac{h_t}{2} \big(\sigma^2 i^2 + ri\big) U_{i+1,j}, \end{aligned}} \tag{33}$$

for $i = 1, \dots, N_S - 1$ and $j = 0, \dots, N_t - 1$.

> Interpretation
>
> The explicit method computes $U_{i,j+1}$ directly from known values at time level j.

4.1 Stability analysis

We now analyse the stability of the explicit scheme

$$U_{i,j+1} = a_i U_{i-1,j} + b_i U_{i,j} + c_i U_{i+1,j}, \tag{34}$$

where

$$a_i = \frac{h_t}{2}\big(\sigma^2 i^2 - ri\big), \qquad b_i = 1 - \sigma^2 i^2 h_t - r h_t, \qquad c_i = \frac{h_t}{2}\big(\sigma^2 i^2 + ri\big). \tag{35}$$

For the heat equation, the Von Neumann analysis can be applied directly because the coefficients of the finite-difference scheme are constant. In the Black–Scholes equation, however, the coefficient multiplying the second derivative is $\frac{\sigma^2}{2}S^2$, and the coefficient multiplying the first derivative is rS. Therefore, after discretisation on the uniform grid $S_i = ih_S$, the coefficients depend on the spatial index i.

For this reason, we perform a local Von Neumann analysis: at a fixed spatial node i, we freeze the coefficients a_i, b_i, c_i and study the behaviour of a Fourier mode

$$U_{i,j} = R_j e^{\mathrm{i}\theta i}, \tag{36}$$

where $\mathrm{i} = \sqrt{-1}$ and θ is the dimensionless wave number.

Substituting this expression into (34) gives

$$R_{j+1} e^{\mathrm{i}\theta i} = a_i R_j e^{\mathrm{i}\theta(i-1)} + b_i R_j e^{\mathrm{i}\theta i} + c_i R_j e^{\mathrm{i}\theta(i+1)}. \tag{37}$$

Dividing by $R_j e^{\mathrm{i}\theta i}$, we obtain the amplification factor

$$G_i(\theta) := \frac{R_{j+1}}{R_j} = a_i e^{-\mathrm{i}\theta} + b_i + c_i e^{\mathrm{i}\theta}. \tag{38}$$

Using

$$e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta} = 2\cos\theta, \qquad e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta} = 2\mathrm{i}\sin\theta,$$

we can write

$$G_i(\theta) = b_i + (a_i + c_i)\cos\theta + \mathrm{i}(c_i - a_i)\sin\theta. \tag{39}$$

Since

$$a_i + c_i = \sigma^2 i^2 h_t, \qquad c_i - a_i = r i h_t,$$

we get

$$G_i(\theta) = 1 - \sigma^2 i^2 h_t - r h_t + \sigma^2 i^2 h_t \cos\theta + \mathrm{i} r i h_t \sin\theta. \tag{40}$$

Equivalently, using $1 - \cos\theta = 2\sin^2(\theta/2)$,

$$G_i(\theta) = 1 - r h_t - 2\sigma^2 i^2 h_t \sin^2\left(\frac{\theta}{2}\right) + \mathrm{i} r i h_t \sin\theta. \tag{41}$$

The Von Neumann stability condition requires

$$|G_i(\theta)| \leq 1 \tag{42}$$

for every wave number θ and for every interior node $i = 1, \dots, N_S - 1$. Therefore, from (41),

$$\left[1 - rh_t - 2\sigma^2 i^2 h_t \sin^2\left(\frac{\theta}{2}\right)\right]^2 + [rih_t \sin\theta]^2 \leq 1. \tag{43}$$

This is the local Von Neumann stability condition for the centered explicit Black–Scholes scheme. Unlike the heat equation condition, it is not a simple single restriction on h_t/h_S^2, because the amplification factor contains both the diffusion contribution and the drift contribution.

A simple sufficient condition is obtained by requiring the explicit update (34) to be monotone, that is, by requiring all weights to be non-negative:

$$a_i \geq 0, \qquad b_i \geq 0, \qquad c_i \geq 0, \qquad i = 1, \dots, N_S - 1. \tag{44}$$

This condition is stronger than the exact Von Neumann condition (43), but it is easier to interpret. If the weights are non-negative, then

$$|G_i(\theta)| = \left|a_i e^{-\mathrm{i}\theta} + b_i + c_i e^{\mathrm{i}\theta}\right| \tag{45}$$

$$\leq a_i + b_i + c_i. \tag{46}$$

But

$$a_i + b_i + c_i = 1 - rh_t. \tag{47}$$

Therefore, if $a_i, b_i, c_i \geq 0$, then

$$|G_i(\theta)| \leq 1 - rh_t \leq 1, \tag{48}$$

provided $h_t > 0$ and $rh_t \leq 1$. Hence, the non-negativity of the explicit weights gives a practical sufficient stability condition.

We now derive these conditions. First,

$$a_i = \frac{h_t}{2}(\sigma^2 i^2 - ri) \geq 0. \tag{49}$$

Since $h_t > 0$, this is equivalent to

$$\sigma^2 i^2 - ri \geq 0. \tag{50}$$

Factoring i, and using $i \geq 1$, gives

$$\sigma^2 i - r \geq 0. \tag{51}$$

The most restrictive case is $i = 1$. Therefore,

$$\boxed{\sigma^2 \geq r.} \tag{52}$$

Second,

$$b_i = 1 - \sigma^2 i^2 h_t - rh_t \geq 0. \tag{53}$$

Thus,

$$1 - h_t(\sigma^2 i^2 + r) \geq 0, \tag{54}$$

or

$$h_t \leq \frac{1}{\sigma^2 i^2 + r}. \tag{55}$$

This must hold for every interior node $i = 1, \ldots, N_S - 1$. Since the denominator is increasing in i, the most restrictive condition occurs at $i = N_S - 1$. Hence,

$$\boxed{h_t \leq \frac{1}{\sigma^2 (N_S - 1)^2 + r}.} \tag{56}$$

Finally,

$$c_i = \frac{h_t}{2}\bigl(\sigma^2 i^2 + ri\bigr) \geq 0 \tag{57}$$

is automatically satisfied for $h_t > 0$, $\sigma > 0$, $r > 0$, and $i \geq 1$.

Therefore, a sufficient stability and monotonicity condition for the centered explicit scheme is

$$\boxed{\sigma^2 \geq r, \qquad h_t \leq \frac{1}{\sigma^2 (N_S - 1)^2 + r}.} \tag{58}$$

Interpretation

The explicit Black–Scholes scheme is conditionally stable. The restriction on h_t becomes more severe as N_S increases, because the coefficient $\sigma^2 i^2$ is largest near the upper boundary $S = S_{\max}$. This is the Black–Scholes analogue of the heat equation restriction $\lambda = h_t/h_S^2 \leq 1/2$, but with spatially dependent coefficients.

5 Implicit method

The implicit method evaluates the spatial finite differences at the unknown time level $j + 1$:

$$\frac{U_{i,j+1} - U_{i,j}}{h_t} = \frac{\sigma^2}{2} S_i^2 \left(\frac{U_{i+1,j+1} - 2U_{i,j+1} + U_{i-1,j+1}}{h_S^2} \right) + rS_i \left(\frac{U_{i+1,j+1} - U_{i-1,j+1}}{2h_S} \right) - rU_{i,j+1}. \tag{59}$$

Multiplying by h_t and collecting unknowns on the left-hand side gives

$$\boxed{\begin{aligned} &-\frac{h_t}{2}\bigl(\sigma^2 i^2 - ri\bigr) U_{i-1,j+1} \\ &+ \bigl(1 + \sigma^2 i^2 h_t + rh_t\bigr) U_{i,j+1} \\ &-\frac{h_t}{2}\bigl(\sigma^2 i^2 + ri\bigr) U_{i+1,j+1} = U_{i,j}. \end{aligned}} \tag{60}$$

Equivalently, define

$$\boxed{\begin{aligned} \alpha_i^I &= -\frac{h_t}{2}\bigl(\sigma^2 i^2 - ri\bigr), \\ \beta_i^I &= 1 + \sigma^2 i^2 h_t + rh_t, \\ \gamma_i^I &= -\frac{h_t}{2}\bigl(\sigma^2 i^2 + ri\bigr). \end{aligned}} \tag{61}$$

Then

$$\boxed{\alpha_i^I U_{i-1,j+1} + \beta_i^I U_{i,j+1} + \gamma_i^I U_{i+1,j+1} = U_{i,j}.} \tag{62}$$

Interpretation

The implicit method requires solving a tridiagonal linear system at each time step.

Implicit method in matrix form

Let

$$\mathbf{U}^{j+1} = \begin{pmatrix} U_{1,j+1} \\ U_{2,j+1} \\ \vdots \\ U_{N_S-1,j+1} \end{pmatrix}. \tag{63}$$

Then

$$\boxed{A_I \mathbf{U}^{j+1} = \mathbf{U}^j + \mathbf{q}_I^{j+1},} \tag{64}$$

where

$$A_I = \begin{pmatrix} \beta_1^I & \gamma_1^I & 0 & \cdots & 0 \\ \alpha_2^I & \beta_2^I & \gamma_2^I & \ddots & \vdots \\ 0 & \alpha_3^I & \beta_3^I & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \gamma_{N_S-2}^I \\ 0 & \cdots & 0 & \alpha_{N_S-1}^I & \beta_{N_S-1}^I \end{pmatrix}, \tag{65}$$

and

$$\mathbf{q}_I^{j+1} = \begin{pmatrix} -\alpha_1^I U_{0,j+1} \\ 0 \\ \vdots \\ 0 \\ -\gamma_{N_S-1}^I U_{N_S,j+1} \end{pmatrix}. \tag{66}$$

5.1 Stability analysis

We now analyse the stability of the implicit scheme using a local Von Neumann argument. The word local is important because, on the uniform grid $S_i = ih_S$, the Black–Scholes coefficients depend on the spatial index i. Thus, at a fixed node i, we freeze the coefficients and study the amplification of a Fourier mode.

Starting from

$$\alpha_i^I U_{i-1,j+1} + \beta_i^I U_{i,j+1} + \gamma_i^I U_{i+1,j+1} = U_{i,j}, \tag{67}$$

we take

$$U_{i,j} = R_j e^{\mathrm{i}\theta i}, \tag{68}$$

where $\mathrm{i} = \sqrt{-1}$ and θ is the dimensionless wave number.

Substituting into the implicit scheme gives

$$\alpha_i^I R_{j+1} e^{\mathrm{i}\theta(i-1)} + \beta_i^I R_{j+1} e^{\mathrm{i}\theta i} + \gamma_i^I R_{j+1} e^{\mathrm{i}\theta(i+1)} = R_j e^{\mathrm{i}\theta i}. \tag{69}$$

Dividing by $e^{\mathrm{i}\theta i}$, we obtain

$$R_{j+1}\left(\alpha_i^I e^{-\mathrm{i}\theta} + \beta_i^I + \gamma_i^I e^{\mathrm{i}\theta}\right) = R_j. \tag{70}$$

Hence the amplification factor is

$$G_i(\theta) := \frac{R_{j+1}}{R_j} = \frac{1}{\alpha_i^I e^{-\mathrm{i}\theta} + \beta_i^I + \gamma_i^I e^{\mathrm{i}\theta}}. \tag{71}$$

Using

$$e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta} = 2\cos\theta, \qquad e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta} = 2\mathrm{i}\sin\theta,$$

the denominator can be written as

$$\alpha_i^I e^{-\mathrm{i}\theta} + \beta_i^I + \gamma_i^I e^{\mathrm{i}\theta} = \beta_i^I + (\alpha_i^I + \gamma_i^I)\cos\theta + \mathrm{i}(\gamma_i^I - \alpha_i^I)\sin\theta. \tag{72}$$

From the definitions of the implicit coefficients,

$$\alpha_i^I + \gamma_i^I = -\sigma^2 i^2 h_t, \qquad \gamma_i^I - \alpha_i^I = -rih_t. \tag{73}$$

Therefore,

$$\begin{aligned}
\alpha_i^I e^{-\mathrm{i}\theta} + \beta_i^I + \gamma_i^I e^{\mathrm{i}\theta} &= 1 + \sigma^2 i^2 h_t + rh_t - \sigma^2 i^2 h_t \cos\theta - \mathrm{i}rih_t \sin\theta & (74)\\
&= 1 + rh_t + \sigma^2 i^2 h_t (1 - \cos\theta) - \mathrm{i}rih_t \sin\theta. & (75)
\end{aligned}$$

Using $1 - \cos\theta = 2\sin^2(\theta/2)$, this becomes

$$\alpha_i^I e^{-\mathrm{i}\theta} + \beta_i^I + \gamma_i^I e^{\mathrm{i}\theta} = 1 + rh_t + 2\sigma^2 i^2 h_t \sin^2\left(\frac{\theta}{2}\right) - \mathrm{i}rih_t \sin\theta. \tag{76}$$

Hence

$$G_i(\theta) = \frac{1}{1 + rh_t + 2\sigma^2 i^2 h_t \sin^2\left(\frac{\theta}{2}\right) - \mathrm{i}rih_t \sin\theta}. \tag{77}$$

The Von Neumann stability condition requires

$$|G_i(\theta)| \le 1 \tag{78}$$

for every wave number θ and for every interior node $i = 1, \ldots, N_S - 1$. From (77), we have

$$|G_i(\theta)|^2 = \frac{1}{\left[1 + rh_t + 2\sigma^2 i^2 h_t \sin^2\left(\frac{\theta}{2}\right)\right]^2 + [rih_t \sin\theta]^2}. \tag{79}$$

Assuming $r \ge 0$, $\sigma > 0$, and $h_t > 0$, the first term in the denominator satisfies

$$1 + rh_t + 2\sigma^2 i^2 h_t \sin^2\left(\frac{\theta}{2}\right) \ge 1, \tag{80}$$

and the second term is non-negative:

$$[rih_t \sin\theta]^2 \ge 0. \tag{81}$$

Therefore the denominator is always greater than or equal to 1, and so

$$|G_i(\theta)|^2 \le 1. \tag{82}$$

Equivalently,

$$\boxed{|G_i(\theta)| \le 1.} \tag{83}$$

This holds for every θ, every interior node i, and every choice of $h_t > 0$. Therefore, the implicit method is unconditionally stable in the Von Neumann sense.

Interpretation

Unlike the explicit method, the implicit method does not require a restriction of the form $h_t \leq Ch_S^2$. The spatial operator is evaluated at the unknown time level $j+1$, so the amplification factor appears in the denominator. As a result, high-frequency modes are damped rather than amplified, independently of the relation between the time step h_t and the spatial step h_S.

6 Crank–Nicolson method

The Crank–Nicolson method averages the full right-hand side of the Black–Scholes equation between time levels j and $j+1$:

$$\begin{aligned}\frac{U_{i,j+1}-U_{i,j}}{h_t} &= \frac{1}{2}\left[\frac{\sigma^2}{2}S_i^2\left(\frac{U_{i+1,j}-2U_{i,j}+U_{i-1,j}}{h_S^2}\right)+rS_i\left(\frac{U_{i+1,j}-U_{i-1,j}}{2h_S}\right)-rU_{i,j}\right]\\ &\quad+\frac{1}{2}\left[\frac{\sigma^2}{2}S_i^2\left(\frac{U_{i+1,j+1}-2U_{i,j+1}+U_{i-1,j+1}}{h_S^2}\right)+rS_i\left(\frac{U_{i+1,j+1}-U_{i-1,j+1}}{2h_S}\right)-rU_{i,j+1}\right].\end{aligned} \tag{84}$$

Multiplying by h_t, collecting the $j+1$ terms on the left and the j terms on the right gives

$$\begin{aligned}&U_{i-1,j+1}\left(-\frac{\sigma^2}{4}h_t\frac{S_i^2}{h_S^2}+\frac{r}{4}h_t\frac{S_i}{h_S}\right)+U_{i,j+1}\left(1+\frac{\sigma^2}{2}h_t\frac{S_i^2}{h_S^2}+\frac{rh_t}{2}\right)\\ &\quad+U_{i+1,j+1}\left(-\frac{\sigma^2}{4}h_t\frac{S_i^2}{h_S^2}-\frac{r}{4}h_t\frac{S_i}{h_S}\right)\\ &=U_{i-1,j}\left(\frac{\sigma^2}{4}h_t\frac{S_i^2}{h_S^2}-\frac{r}{4}h_t\frac{S_i}{h_S}\right)+U_{i,j}\left(1-\frac{\sigma^2}{2}h_t\frac{S_i^2}{h_S^2}-\frac{rh_t}{2}\right)\\ &\quad+U_{i+1,j}\left(\frac{\sigma^2}{4}h_t\frac{S_i^2}{h_S^2}+\frac{r}{4}h_t\frac{S_i}{h_S}\right).\end{aligned} \tag{85}$$

Using $S_i/h_S = i$, define

$$\boxed{\begin{aligned}a_i &= -\frac{\sigma^2}{4}h_t i^2+\frac{rh_t}{4}i,\\ b_i &= 1+\frac{\sigma^2}{2}h_t i^2+\frac{rh_t}{2},\\ c_i &= -\frac{\sigma^2}{4}h_t i^2-\frac{rh_t}{4}i,\\ d_i &= 1-\frac{\sigma^2}{2}h_t i^2-\frac{rh_t}{2}.\end{aligned}} \tag{86}$$

Therefore,

$$\boxed{a_iU_{i-1,j+1}+b_iU_{i,j+1}+c_iU_{i+1,j+1}=-a_iU_{i-1,j}+d_iU_{i,j}-c_iU_{i+1,j}.} \tag{87}$$

Crank–Nicolson matrix form

Let

$$\mathbf{U}^j = \begin{pmatrix} U_{1,j} \\ U_{2,j} \\ \vdots \\ U_{N_S-1,j} \end{pmatrix}. \tag{88}$$

Define

$$A_{CN} = \begin{pmatrix} b_1 & c_1 & 0 & \cdots & 0 \\ a_2 & b_2 & c_2 & \ddots & \vdots \\ 0 & a_3 & b_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & c_{N_S-2} \\ 0 & \cdots & 0 & a_{N_S-1} & b_{N_S-1} \end{pmatrix}, \tag{89}$$

$$B_{CN} = \begin{pmatrix} d_1 & -c_1 & 0 & \cdots & 0 \\ -a_2 & d_2 & -c_2 & \ddots & \vdots \\ 0 & -a_3 & d_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -c_{N_S-2} \\ 0 & \cdots & 0 & -a_{N_S-1} & d_{N_S-1} \end{pmatrix}. \tag{90}$$

Then

$$\boxed{A_{CN}\mathbf{U}^{j+1} = B_{CN}\mathbf{U}^j + \mathbf{q}_{CN}^{j,j+1},} \tag{91}$$

with

$$\boxed{\mathbf{q}_{CN}^{j,j+1} = \begin{pmatrix} -a_1 U_{0,j+1} - a_1 U_{0,j} \\ 0 \\ \vdots \\ 0 \\ -c_{N_S-1}U_{N_S,j+1} - c_{N_S-1}U_{N_S,j} \end{pmatrix}.} \tag{92}$$

6.1 Order of accuracy

We now justify why the Crank–Nicolson method is second order in space and second order in time.

The scheme uses the centered finite-difference approximation for the first spatial derivative:

$$\frac{\partial U}{\partial S}(S_i, t_j) = \frac{U_{i+1,j} - U_{i-1,j}}{2h_S} + O(h_S^2), \tag{93}$$

and the centered finite-difference approximation for the second spatial derivative:

$$\frac{\partial^2 U}{\partial S^2}(S_i, t_j) = \frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{h_S^2} + O(h_S^2). \tag{94}$$

The same approximations are used at time level $j+1$:

$$\frac{\partial U}{\partial S}(S_i, t_{j+1}) = \frac{U_{i+1,j+1} - U_{i-1,j+1}}{2h_S} + O(h_S^2), \tag{95}$$

and

$$\frac{\partial^2 U}{\partial S^2}(S_i, t_{j+1}) = \frac{U_{i+1,j+1} - 2U_{i,j+1} + U_{i-1,j+1}}{h_S^2} + O(h_S^2). \tag{96}$$

Therefore, at each of the two time levels j and $j+1$, the spatial part of the Black–Scholes equation is approximated with error $O(h_S^2)$. Since Crank–Nicolson takes the average of these two spatial approximations, the spatial error remains

$$O(h_S^2). \tag{97}$$

We now analyse the time discretisation. Expanding $U(S_i, t_{j+1})$ around t_j, with S_i fixed, gives

$$U(S_i, t_{j+1}) = U(S_i, t_j) + h_t \frac{\partial U}{\partial t}(S_i, t_j) + \frac{h_t^2}{2}\frac{\partial^2 U}{\partial t^2}(S_i, t_j) + \frac{h_t^3}{6}\frac{\partial^3 U}{\partial t^3}(S_i, t_j) + O(h_t^4). \tag{98}$$

Dividing by h_t, we obtain

$$\frac{U(S_i, t_{j+1}) - U(S_i, t_j)}{h_t} = \frac{\partial U}{\partial t}(S_i, t_j) + \frac{h_t}{2}\frac{\partial^2 U}{\partial t^2}(S_i, t_j) + \frac{h_t^2}{6}\frac{\partial^3 U}{\partial t^3}(S_i, t_j) + O(h_t^3). \tag{99}$$

On the other hand, Crank–Nicolson does not use only the derivative at t_j. It averages the Black–Scholes right-hand side between t_j and t_{j+1}. For the exact solution, the Black–Scholes equation gives

$$\frac{\partial U}{\partial t}(S_i, t_j) = \frac{\sigma^2}{2} S_i^2 \frac{\partial^2 U}{\partial S^2}(S_i, t_j) + rS_i \frac{\partial U}{\partial S}(S_i, t_j) - rU(S_i, t_j), \tag{100}$$

and

$$\frac{\partial U}{\partial t}(S_i, t_{j+1}) = \frac{\sigma^2}{2} S_i^2 \frac{\partial^2 U}{\partial S^2}(S_i, t_{j+1}) + rS_i \frac{\partial U}{\partial S}(S_i, t_{j+1}) - rU(S_i, t_{j+1}). \tag{101}$$

Therefore, before replacing the spatial derivatives by finite differences, the Crank–Nicolson time average corresponds to

$$\frac{1}{2}\left[\frac{\partial U}{\partial t}(S_i, t_j) + \frac{\partial U}{\partial t}(S_i, t_{j+1})\right]. \tag{102}$$

Expanding $\frac{\partial U}{\partial t}(S_i, t_{j+1})$ around t_j, we get

$$\frac{\partial U}{\partial t}(S_i, t_{j+1}) = \frac{\partial U}{\partial t}(S_i, t_j) + h_t \frac{\partial^2 U}{\partial t^2}(S_i, t_j) + \frac{h_t^2}{2}\frac{\partial^3 U}{\partial t^3}(S_i, t_j) + O(h_t^3). \tag{103}$$

Hence,

$$\begin{aligned}\frac{1}{2}\left[\frac{\partial U}{\partial t}(S_i, t_j) + \frac{\partial U}{\partial t}(S_i, t_{j+1})\right] &= \frac{\partial U}{\partial t}(S_i, t_j) + \frac{h_t}{2}\frac{\partial^2 U}{\partial t^2}(S_i, t_j) \\ &\quad + \frac{h_t^2}{4}\frac{\partial^3 U}{\partial t^3}(S_i, t_j) + O(h_t^3).\end{aligned} \tag{104}$$

Comparing (99) and (104), the terms

$$\frac{\partial U}{\partial t}(S_i, t_j) \qquad \text{and} \qquad \frac{h_t}{2}\frac{\partial^2 U}{\partial t^2}(S_i, t_j)$$

are the same. The first difference appears only in the terms of order h_t^2:

$$\frac{h_t^2}{6}\frac{\partial^3 U}{\partial t^3}(S_i, t_j) - \frac{h_t^2}{4}\frac{\partial^3 U}{\partial t^3}(S_i, t_j) = -\frac{h_t^2}{12}\frac{\partial^3 U}{\partial t^3}(S_i, t_j). \tag{105}$$

Thus the time discretisation error is

$$O(h_t^2). \tag{106}$$

Combining the spatial and temporal contributions, the Crank–Nicolson method has local truncation error

$$\boxed{O(h_t^2) + O(h_S^2).} \tag{107}$$

Therefore,

$$\boxed{\text{Crank–Nicolson is second order in time and second order in space.}} \tag{108}$$

Interpretation

The explicit and implicit methods use one-sided time approximations and are only first order in time. Crank–Nicolson averages the equation between t_j and t_{j+1}, so the first-order time error cancels. The method remains second order in space because the first and second spatial derivatives are both approximated with centered finite differences.

6.2 Stability analysis

We now analyse the stability of the Crank–Nicolson scheme using a local Von Neumann argument. As in the explicit and implicit cases, the word local is important because the Black–Scholes coefficients depend on the spatial index i through $S_i = ih_S$. Thus, at a fixed node i, we freeze the coefficients and study the amplification of a Fourier mode.

Starting from

$$a_i U_{i-1,j+1} + b_i U_{i,j+1} + c_i U_{i+1,j+1} = -a_i U_{i-1,j} + d_i U_{i,j} - c_i U_{i+1,j}, \tag{109}$$

we take

$$U_{i,j} = R_j e^{\mathrm{i}\theta i}, \tag{110}$$

where $\mathrm{i} = \sqrt{-1}$ and θ is the dimensionless wave number.

Substituting this Fourier mode into the Crank–Nicolson scheme gives

$$\begin{aligned} a_i R_{j+1} e^{\mathrm{i}\theta(i-1)} + b_i R_{j+1} e^{\mathrm{i}\theta i} + c_i R_{j+1} e^{\mathrm{i}\theta(i+1)} \\ = -a_i R_j e^{\mathrm{i}\theta(i-1)} + d_i R_j e^{\mathrm{i}\theta i} - c_i R_j e^{\mathrm{i}\theta(i+1)}. \end{aligned} \tag{111}$$

Dividing by $e^{\mathrm{i}\theta i}$, we obtain

$$R_{j+1}\left(a_i e^{-\mathrm{i}\theta} + b_i + c_i e^{\mathrm{i}\theta}\right) = R_j\left(-a_i e^{-\mathrm{i}\theta} + d_i - c_i e^{\mathrm{i}\theta}\right). \tag{112}$$

Hence the amplification factor is

$$G_i(\theta) := \frac{R_{j+1}}{R_j} = \frac{-a_i e^{-\mathrm{i}\theta} + d_i - c_i e^{\mathrm{i}\theta}}{a_i e^{-\mathrm{i}\theta} + b_i + c_i e^{\mathrm{i}\theta}}. \tag{113}$$

We now simplify numerator and denominator separately. Using

$$e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta} = 2\cos\theta, \qquad e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta} = 2\mathrm{i}\sin\theta,$$

the denominator is

$$a_i e^{-\mathrm{i}\theta} + b_i + c_i e^{\mathrm{i}\theta} = b_i + (a_i + c_i)\cos\theta + \mathrm{i}(c_i - a_i)\sin\theta. \tag{114}$$

From the definitions of a_i, b_i, c_i, d_i,

$$a_i + c_i = -\frac{\sigma^2}{2} h_t i^2, \qquad c_i - a_i = -\frac{r h_t}{2} i. \tag{115}$$

Therefore,

$$\begin{aligned} a_i e^{-\mathrm{i}\theta} + b_i + c_i e^{\mathrm{i}\theta} &= 1 + \frac{\sigma^2}{2} h_t i^2 + \frac{r h_t}{2} - \frac{\sigma^2}{2} h_t i^2 \cos\theta - \mathrm{i}\frac{r h_t}{2} i \sin\theta & (116) \\ &= 1 + \frac{r h_t}{2} + \frac{\sigma^2}{2} h_t i^2 (1 - \cos\theta) - \mathrm{i}\frac{r h_t}{2} i \sin\theta. & (117) \end{aligned}$$

Using $1 - \cos\theta = 2\sin^2(\theta/2)$, this becomes

$$a_i e^{-\mathrm{i}\theta} + b_i + c_i e^{\mathrm{i}\theta} = 1 + \frac{r h_t}{2} + \sigma^2 h_t i^2 \sin^2\left(\frac{\theta}{2}\right) - \mathrm{i}\frac{r h_t}{2} i \sin\theta. \tag{118}$$

Similarly, the numerator is

$$-a_i e^{-\mathrm{i}\theta} + d_i - c_i e^{\mathrm{i}\theta} = d_i - (a_i + c_i)\cos\theta + \mathrm{i}(a_i - c_i)\sin\theta. \tag{119}$$

Since

$$a_i - c_i = \frac{r h_t}{2} i, \tag{120}$$

we get

$$\begin{aligned} -a_i e^{-\mathrm{i}\theta} + d_i - c_i e^{\mathrm{i}\theta} &= 1 - \frac{\sigma^2}{2} h_t i^2 - \frac{r h_t}{2} + \frac{\sigma^2}{2} h_t i^2 \cos\theta + \mathrm{i}\frac{r h_t}{2} i \sin\theta & (121) \\ &= 1 - \frac{r h_t}{2} - \frac{\sigma^2}{2} h_t i^2 (1 - \cos\theta) + \mathrm{i}\frac{r h_t}{2} i \sin\theta. & (122) \end{aligned}$$

Again using $1 - \cos\theta = 2\sin^2(\theta/2)$,

$$-a_i e^{-\mathrm{i}\theta} + d_i - c_i e^{\mathrm{i}\theta} = 1 - \frac{r h_t}{2} - \sigma^2 h_t i^2 \sin^2\left(\frac{\theta}{2}\right) + \mathrm{i}\frac{r h_t}{2} i \sin\theta. \tag{123}$$

Define

$$P_i(\theta) = \frac{r h_t}{2} + \sigma^2 h_t i^2 \sin^2\left(\frac{\theta}{2}\right), \qquad Q_i(\theta) = \frac{r h_t}{2} i \sin\theta. \tag{124}$$

Then (118) and (123) can be written compactly as

$$a_i e^{-\mathrm{i}\theta} + b_i + c_i e^{\mathrm{i}\theta} = 1 + P_i(\theta) - \mathrm{i}Q_i(\theta), \tag{125}$$

and

$$-a_i e^{-\mathrm{i}\theta} + d_i - c_i e^{\mathrm{i}\theta} = 1 - P_i(\theta) + \mathrm{i}Q_i(\theta). \tag{126}$$

Therefore, the Crank–Nicolson amplification factor is

$$G_i(\theta) = \frac{1 - P_i(\theta) + \mathrm{i}Q_i(\theta)}{1 + P_i(\theta) - \mathrm{i}Q_i(\theta)}. \tag{127}$$

The Von Neumann stability condition requires

$$|G_i(\theta)| \leq 1 \tag{128}$$

for every wave number θ and for every interior node $i = 1, \ldots, N_S - 1$. From (127),

$$|G_i(\theta)|^2 = \frac{[1 - P_i(\theta)]^2 + [Q_i(\theta)]^2}{[1 + P_i(\theta)]^2 + [Q_i(\theta)]^2}. \tag{129}$$

Assuming $r \geq 0$, $\sigma > 0$, and $h_t > 0$, we have

$$P_i(\theta) = \frac{rh_t}{2} + \sigma^2 h_t i^2 \sin^2\left(\frac{\theta}{2}\right) \geq 0. \tag{130}$$

Hence,

$$[1 + P_i(\theta)]^2 - [1 - P_i(\theta)]^2 = 4P_i(\theta) \geq 0. \tag{131}$$

Therefore,

$$[1 - P_i(\theta)]^2 + [Q_i(\theta)]^2 \leq [1 + P_i(\theta)]^2 + [Q_i(\theta)]^2. \tag{132}$$

From (129), it follows that

$$|G_i(\theta)|^2 \leq 1. \tag{133}$$

Equivalently,

$$\boxed{|G_i(\theta)| \leq 1.} \tag{134}$$

This holds for every wave number θ, every interior node i, and every choice of $h_t > 0$. Therefore, the Crank–Nicolson method is unconditionally stable in the Von Neumann sense.

Interpretation

The Crank–Nicolson method is unconditionally stable because its amplification factor has the form

$$G_i(\theta) = \frac{1 - P_i(\theta) + \mathrm{i}Q_i(\theta)}{1 + P_i(\theta) - \mathrm{i}Q_i(\theta)}, \qquad P_i(\theta) \geq 0.$$

The denominator is always at least as large as the numerator in modulus, so no restriction relating h_t and h_S is needed for stability.

Remark

Unconditional stability does not mean that arbitrarily large time steps give accurate solutions. Crank–Nicolson is stable for any $h_t > 0$, but accuracy still requires sufficiently fine time and space discretisations. Moreover, for very large time steps, high-frequency modes may be weakly damped or oscillatory, even though their modulus remains bounded by one.

7 Method of Lines with Runge–Kutta

The Method of Lines discretises the spatial variable first, while keeping time continuous. Therefore, instead of obtaining a fully discrete scheme immediately, we first transform the Black–Scholes PDE into a system of ordinary differential equations in time.

Starting from

$$\frac{\partial U}{\partial t} = \frac{\sigma^2}{2} S^2 \frac{\partial^2 U}{\partial S^2} + rS\frac{\partial U}{\partial S} - rU, \tag{135}$$

we define the spatial grid

$$S_i = ih_S, \qquad i = 0, \dots, N_S, \qquad h_S = \frac{S_{\max}}{N_S}. \tag{136}$$

At this stage, time is not yet discretised. We write

$$U_i(t) \approx U(S_i, t), \qquad i = 0, \dots, N_S. \tag{137}$$

The boundary values are known:

$$U_0(t) = U(S_0, t) = u_a(t), \qquad U_{N_S}(t) = U(S_{N_S}, t) = u_b(t). \tag{138}$$

For the interior nodes $i = 1, \dots, N_S - 1$, we approximate the spatial derivatives using centered finite differences:

$$\frac{\partial U}{\partial S}(S_i, t) = \frac{U_{i+1}(t) - U_{i-1}(t)}{2h_S} + O(h_S^2), \tag{139}$$

and

$$\frac{\partial^2 U}{\partial S^2}(S_i, t) = \frac{U_{i+1}(t) - 2U_i(t) + U_{i-1}(t)}{h_S^2} + O(h_S^2). \tag{140}$$

Neglecting the truncation errors, the semi-discrete equation is

$$\begin{aligned} \frac{dU_i}{dt}(t) = {} & \frac{\sigma^2}{2} S_i^2 \left(\frac{U_{i+1}(t) - 2U_i(t) + U_{i-1}(t)}{h_S^2} \right) \\ & + rS_i \left(\frac{U_{i+1}(t) - U_{i-1}(t)}{2h_S} \right) - rU_i(t). \end{aligned} \tag{141}$$

Using

$$\frac{S_i}{h_S} = i, \qquad \frac{S_i^2}{h_S^2} = i^2, \tag{142}$$

we obtain

$$\begin{aligned} \frac{dU_i}{dt}(t) = {} & \left(\frac{\sigma^2}{2} i^2 - \frac{r}{2} i \right) U_{i-1}(t) + \left(-\sigma^2 i^2 - r \right) U_i(t) \\ & + \left(\frac{\sigma^2}{2} i^2 + \frac{r}{2} i \right) U_{i+1}(t). \end{aligned} \tag{143}$$

Define

$$\boxed{\begin{aligned} \alpha_i^{ML} &= \frac{\sigma^2}{2} i^2 - \frac{r}{2} i, \\ \beta_i^{ML} &= -\sigma^2 i^2 - r, \\ \gamma_i^{ML} &= \frac{\sigma^2}{2} i^2 + \frac{r}{2} i. \end{aligned}} \tag{144}$$

Then

$$\boxed{\frac{dU_i}{dt}(t) = \alpha_i^{ML} U_{i-1}(t) + \beta_i^{ML} U_i(t) + \gamma_i^{ML} U_{i+1}(t),} \tag{145}$$

for

$$i = 1, \dots, N_S - 1. \tag{146}$$

Interpretation

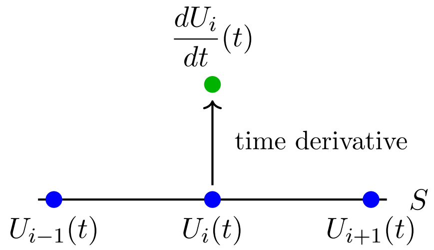
The Method of Lines converts the PDE into a system of ODEs. The unknowns are the time-dependent functions $U_i(t)$, one for each interior spatial grid point. Time is still continuous at this stage.

Method of Lines spatial stencil

The Method of Lines stencil is a spatial stencil. The three values

$$U_{i-1}(t), \qquad U_i(t), \qquad U_{i+1}(t)$$

are all evaluated at the same continuous time t. They define the time derivative of the centre component, $\frac{dU_i}{dt}(t)$. No time step has yet been taken.



$$\frac{dU_i}{dt}(t) = \alpha_i^{ML} U_{i-1}(t) + \beta_i^{ML} U_i(t) + \gamma_i^{ML} U_{i+1}(t). \tag{147}$$

Stencil interpretation

The horizontal points are spatial neighbours. The vertical point represents the time derivative generated at the centre node. It is not $U_i(t + h_t)$.

Matrix form

Define the vector of interior unknowns:

$$\mathbf{U}(t) = \begin{pmatrix} U_1(t) \\ U_2(t) \\ \vdots \\ U_{N_S-1}(t) \end{pmatrix}. \tag{148}$$

The first interior equation contains the left boundary value:

$$\frac{dU_1}{dt}(t) = \alpha_1^{ML} u_a(t) + \beta_1^{ML} U_1(t) + \gamma_1^{ML} U_2(t). \tag{149}$$

The last interior equation contains the right boundary value:

$$\frac{dU_{N_S-1}}{dt}(t) = \alpha_{N_S-1}^{ML} U_{N_S-2}(t) + \beta_{N_S-1}^{ML} U_{N_S-1}(t) + \gamma_{N_S-1}^{ML} u_b(t). \tag{150}$$

Therefore, the semi-discrete system can be written as

$$\boxed{\frac{d\mathbf{U}}{dt}(t) = A_{ML}\mathbf{U}(t) + \mathbf{b}_{ML}(t).} \tag{151}$$

The tridiagonal matrix is

$$A_{ML} = \begin{pmatrix} \beta_1^{ML} & \gamma_1^{ML} & 0 & \cdots & 0 \\ \alpha_2^{ML} & \beta_2^{ML} & \gamma_2^{ML} & \ddots & \vdots \\ 0 & \alpha_3^{ML} & \beta_3^{ML} & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \gamma_{N_S-2}^{ML} \\ 0 & \cdots & 0 & \alpha_{N_S-1}^{ML} & \beta_{N_S-1}^{ML} \end{pmatrix}. \tag{152}$$

The boundary contribution vector is

$$\boxed{\mathbf{b}_{ML}(t) = \begin{pmatrix} \alpha_1^{ML} u_a(t) \\ 0 \\ \vdots \\ 0 \\ \gamma_{N_S-1}^{ML} u_b(t) \end{pmatrix}}. \tag{153}$$

The initial condition is obtained from the payoff:

$$\boxed{\mathbf{U}(0) = \begin{pmatrix} u_0(S_1) \\ u_0(S_2) \\ \vdots \\ u_0(S_{N_S-1}) \end{pmatrix}}. \tag{154}$$

Where the boundary vector comes from

The vector $\mathbf{b}_{ML}(t)$ appears because the first and last interior equations contain the boundary values $u_a(t)$ and $u_b(t)$. These boundary values are known and are not included in the unknown vector $\mathbf{U}(t)$.

Classical fourth-order Runge–Kutta method

After the Method of Lines spatial discretisation, the Black–Scholes PDE has been converted into the semi-discrete ODE system

$$\frac{d\mathbf{U}}{dt}(t) = A_{ML}\mathbf{U}(t) + \mathbf{b}_{ML}(t), \tag{155}$$

where

$$\mathbf{U}(t) = \begin{pmatrix} U_1(t) \\ U_2(t) \\ \vdots \\ U_{N_S-1}(t) \end{pmatrix}. \tag{156}$$

We now discretise time:

$$t_j = jh_t, \qquad j = 0, \dots, N_t, \qquad h_t = \frac{T}{N_t}. \tag{157}$$

At each time level, the numerical approximation is

$$\mathbf{U}^j \approx \mathbf{U}(t_j), \tag{158}$$

where

$$\mathbf{U}^j = \begin{pmatrix} U_{1,j} \\ U_{2,j} \\ \vdots \\ U_{N_S-1,j} \end{pmatrix}. \tag{159}$$

The classical fourth-order Runge–Kutta method advances the whole vector $\mathbf{U}^j$ to $\mathbf{U}^{j+1}$. It does this by computing four internal stage increments:

$$\boxed{\begin{aligned} \mathbf{k}_1 &= h_t \left(A_{ML}\mathbf{U}^j + \mathbf{b}_{ML}(t_j)\right), \\ \mathbf{k}_2 &= h_t \left[A_{ML}\left(\mathbf{U}^j + \frac{\mathbf{k}_1}{2}\right) + \mathbf{b}_{ML}\left(t_j + \frac{h_t}{2}\right)\right], \\ \mathbf{k}_3 &= h_t \left[A_{ML}\left(\mathbf{U}^j + \frac{\mathbf{k}_2}{2}\right) + \mathbf{b}_{ML}\left(t_j + \frac{h_t}{2}\right)\right], \\ \mathbf{k}_4 &= h_t \left[A_{ML}\left(\mathbf{U}^j + \mathbf{k}_3\right) + \mathbf{b}_{ML}\left(t_j + h_t\right)\right], \\ \mathbf{U}^{j+1} &= \mathbf{U}^j + \frac{1}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right). \end{aligned}} \tag{160}$$

In component form, the updated vector is

$$\mathbf{U}^{j+1} = \begin{pmatrix} U_{1,j+1} \\ U_{2,j+1} \\ \vdots \\ U_{N_S-1,j+1} \end{pmatrix}. \tag{161}$$

Interpretation

The RK4 method does not create four new grid points. It advances the whole interior vector from time level j to time level $j+1$. The quantities $\mathbf{k}_1, \dots, \mathbf{k}_4$ are internal vector increments used to estimate the average slope during the time step.

Runge–Kutta time-step diagram

The RK4 diagram should be read as a time-step diagram, not as a finite-difference stencil. The vertical direction represents time, and the method moves the complete interior vector from $\mathbf{U}^j$ to $\mathbf{U}^{j+1}$.

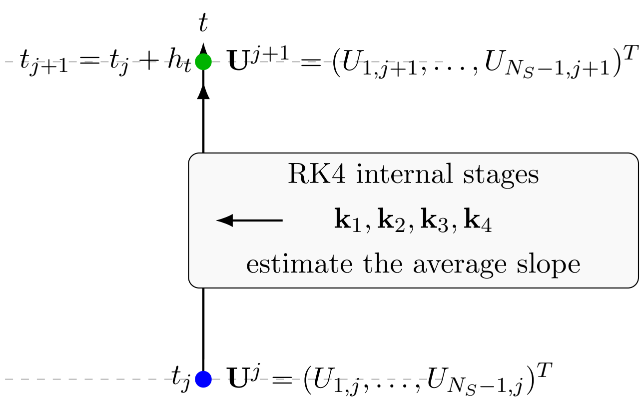


Diagram interpretation

Only $\mathbf{U}^j$ and $\mathbf{U}^{j+1}$ are solution vectors at time levels. The stage increments $\mathbf{k}_1, \ldots, \mathbf{k}_4$ are not solution values and are not spatial positions. They are internal calculations used to form the weighted update.

Order of accuracy

The spatial part uses centered finite differences:

$$\frac{\partial U}{\partial S}(S_i, t) = \frac{U_{i+1}(t) - U_{i-1}(t)}{2h_S} + O(h_S^2), \tag{162}$$

and

$$\frac{\partial^2 U}{\partial S^2}(S_i, t) = \frac{U_{i+1}(t) - 2U_i(t) + U_{i-1}(t)}{h_S^2} + O(h_S^2). \tag{163}$$

Therefore, the spatial discretisation error is

$$O(h_S^2). \tag{164}$$

The time integration is performed with the classical fourth-order Runge–Kutta method, which has temporal error

$$O(h_t^4). \tag{165}$$

Combining both contributions,

$$\boxed{\text{MOL + RK4 accuracy:} \qquad O(h_S^2) + O(h_t^4).} \tag{166}$$

Comparison

Crank–Nicolson is second order in time and second order in space. The Method of Lines with RK4 is fourth order in time and second order in space, because the spatial derivatives are still approximated with centered second-order finite differences.

Stability

The Method of Lines with RK4 uses an explicit time integrator. Therefore, it is not unconditionally stable.

The spatial discretisation produces the ODE system

$$\frac{d\mathbf{U}}{dt}(t) = A_{ML}\mathbf{U}(t) + \mathbf{b}_{ML}(t). \tag{167}$$

The stability of the RK4 time integration depends on the size of h_t relative to the eigenvalues of A_{ML}. Therefore, when the spatial grid is refined, the time step may need to be reduced.

Interpretation

MOL + RK4 has high temporal accuracy, but it remains conditionally stable because RK4 is an explicit time-stepping method.

Recovering the original option value

The PDE was solved in the reversed time variable, so the computed function is $U(S,t)$. The original option value is recovered by

$$\boxed{V(S,t) = U(S,T-t).} \tag{168}$$

At the discrete level, this corresponds to reversing the time direction of the computed solution matrix.

8 American put option and PSOR method

For an American put option, the holder may exercise before maturity. Therefore, the option value must be at least as large as the immediate exercise payoff. In the reversed-time variable

$$\tau = T - t, \qquad U(S,\tau) = V(S, T-\tau), \tag{169}$$

the payoff becomes the initial condition and the obstacle:

$$g(S) = \max(K - S, 0). \tag{170}$$

On the grid

$$S_i = ih_S, \qquad i = 0, \dots, N_S, \qquad h_S = \frac{S_{\max}}{N_S}, \tag{171}$$

we define

$$g_i = \max(K - S_i, 0). \tag{172}$$

For the interior nodes,

$$\mathbf{g} = \begin{pmatrix} g_1 \\ g_2 \\ \vdots \\ g_{N_S-1} \end{pmatrix}. \tag{173}$$

The American constraint is

$$\boxed{U_{i,j} \geq g_i, \qquad i = 1, \dots, N_S - 1.} \tag{174}$$

Financial interpretation

For an American put, the numerical value cannot fall below the immediate exercise payoff. If $U_{i,j} = g_i$, exercise is optimal at that node. If $U_{i,j} > g_i$, continuation is optimal.

Crank–Nicolson system with an obstacle

For the European problem, one Crank–Nicolson time step has the matrix form

$$A_{CN}\mathbf{U}^{j+1} = B_{CN}\mathbf{U}^{j} + \mathbf{q}_{CN}^{j,j+1}. \tag{175}$$

For the American problem, the same matrices are used, but the new vector must also satisfy the obstacle constraint. Define the right-hand side

$$\boxed{\mathbf{r}^{j,j+1} = B_{CN}\mathbf{U}^{j} + \mathbf{q}_{CN}^{j,j+1}.} \tag{176}$$

Without the early-exercise constraint, we would solve

$$A_{CN}\mathbf{U}^{j+1} = \mathbf{r}^{j,j+1}. \tag{177}$$

For the American option, set

$$\mathbf{w} = \mathbf{U}^{j+1}. \tag{178}$$

The new vector must satisfy

$$\mathbf{w} \geq \mathbf{g}. \tag{179}$$

Thus, the time step becomes a linear complementarity problem:

$$\boxed{\begin{aligned} \mathbf{w} - \mathbf{g} &\geq 0, \\ A_{CN}\mathbf{w} - \mathbf{r}^{j,j+1} &\geq 0, \\ (\mathbf{w} - \mathbf{g})^T \left(A_{CN}\mathbf{w} - \mathbf{r}^{j,j+1}\right) &= 0. \end{aligned}} \tag{180}$$

The first inequality enforces the payoff constraint. The second inequality is the discrete Black–Scholes inequality. The last condition means that, at each node, only one of the two alternatives is active: continuation or exercise.

Complementarity interpretation

At a continuation node,

$$w_i > g_i,$$

the obstacle is inactive and the Crank–Nicolson equation holds as an equality. At an exercise node,

$$w_i = g_i,$$

the obstacle is active and the option value is pinned to the payoff.

Shift above the obstacle

Introduce the shifted variable

$$\boxed{\mathbf{x} = \mathbf{w} - \mathbf{g}.} \tag{181}$$

Then

$$\mathbf{x} \geq 0 \tag{182}$$

is exactly the American exercise constraint.

Since

$$\mathbf{w} = \mathbf{x} + \mathbf{g}, \tag{183}$$

we have

$$\begin{aligned} A_{CN}\mathbf{w} - \mathbf{r}^{j,j+1} &= A_{CN}(\mathbf{x} + \mathbf{g}) - \mathbf{r}^{j,j+1} && (184) \\ &= A_{CN}\mathbf{x} - \left(\mathbf{r}^{j,j+1} - A_{CN}\mathbf{g}\right). && (185) \end{aligned}$$

Define

$$\boxed{\widetilde{\mathbf{b}}^{\,j,j+1} = \mathbf{r}^{j,j+1} - A_{CN}\mathbf{g}.} \tag{186}$$

Then the complementarity problem becomes

$$\boxed{\begin{aligned} \mathbf{x} &\geq 0, \\ A_{CN}\mathbf{x} - \widetilde{\mathbf{b}}^{j,j+1} &\geq 0, \\ \mathbf{x}^T\left(A_{CN}\mathbf{x} - \widetilde{\mathbf{b}}^{j,j+1}\right) &= 0. \end{aligned}} \tag{187}$$

Once $\mathbf{x}$ is found, the option value is recovered by

$$\boxed{\mathbf{U}^{j+1} = \mathbf{x} + \mathbf{g}.} \tag{188}$$

Why the shift is useful

The transformation $\mathbf{x} = \mathbf{w} - \mathbf{g}$ changes the lower bound from $\mathbf{w} \geq \mathbf{g}$ to the simpler condition $\mathbf{x} \geq 0$. The PSOR method can then use a SOR step followed by a projection onto the non-negative region.

PSOR componentwise update

Let

$$A_{CN} = D + L + R, \tag{189}$$

where D is the diagonal part, L is the strictly lower triangular part, and R is the strictly upper triangular part.

For a relaxation parameter

$$0 < \omega < 2, \tag{190}$$

the projected SOR iteration updates the components of $\mathbf{x}$ one by one. For $i = 1, \ldots, N_S - 1$, define the tentative SOR value

$$\widehat{x}_i^{(m+1)} = (1-\omega)x_i^{(m)} + \frac{\omega}{(A_{CN})_{ii}}\left[\widetilde{b}_i^{j,j+1} - \sum_{k<i}(A_{CN})_{ik}x_k^{(m+1)} - \sum_{k>i}(A_{CN})_{ik}x_k^{(m)}\right]. \tag{191}$$

Then project:

$$\boxed{x_i^{(m+1)} = \max\left(0, \widehat{x}_i^{(m+1)}\right).} \tag{192}$$

Because A_{CN} is tridiagonal, only the neighbouring components appear:

$$\widehat{x}_i^{(m+1)} = (1-\omega)x_i^{(m)} + \frac{\omega}{b_i}\left[\widetilde{b}_i^{j,j+1} - a_i x_{i-1}^{(m+1)} - c_i x_{i+1}^{(m)}\right], \tag{193}$$

with the missing neighbour omitted at the first and last interior nodes.

Therefore,

$$\boxed{x_i^{(m+1)} = \max\left(0,\ (1-\omega)x_i^{(m)} + \frac{\omega}{b_i}\left[\widetilde{b}_i^{j,j+1} - a_i x_{i-1}^{(m+1)} - c_i x_{i+1}^{(m)}\right]\right).} \tag{194}$$

PSOR interpretation

The SOR part tries to solve the Crank–Nicolson linear system. The projection enforces the American constraint. If the tentative value would place the option below the payoff, the projection resets x_i to zero, which means $U_{i,j+1} = g_i$.

PSOR algorithm at one time step

At a fixed time step $j \to j+1$, the PSOR algorithm is:

1. Compute the Crank–Nicolson right-hand side:
$$\mathbf{r}^{j,j+1} = B_{CN}\mathbf{U}^j + \mathbf{q}_{CN}^{j,j+1}. \tag{195}$$
2. Shift the right-hand side by the obstacle:
$$\widetilde{\mathbf{b}}^{j,j+1} = \mathbf{r}^{j,j+1} - A_{CN}\mathbf{g}. \tag{196}$$
3. Choose an initial guess, for example the previous continuation value:
$$\mathbf{x}^{(0)} = \max\left(\mathbf{U}^j - \mathbf{g}, 0\right). \tag{197}$$
4. Sweep over $i = 1, \dots, N_S - 1$, applying
$$x_i^{(m+1)} = \max\left(0,\ (1-\omega)x_i^{(m)} + \frac{\omega}{b_i}\left[\widetilde{b}_i^{j,j+1} - a_i x_{i-1}^{(m+1)} - c_i x_{i+1}^{(m)}\right]\right). \tag{198}$$
5. Repeat the sweeps until
$$\left\|\mathbf{x}^{(m+1)} - \mathbf{x}^{(m)}\right\|_\infty < \mathrm{tol}, \tag{199}$$
or until a maximum number of iterations is reached.
6. Recover the American option value:
$$\boxed{\mathbf{U}^{j+1} = \mathbf{x}^{(m+1)} + \mathbf{g}.} \tag{200}$$

Algorithmic optimisations

Several computational optimisations follow naturally from the structure of the method.

Use the tridiagonal structure. The matrix A_{CN} is tridiagonal. Therefore, the PSOR sweep should use only the three diagonals a_i, b_i, and c_i. There is no need to store or multiply a full matrix.

Precompute constant quantities. For fixed r, σ, h_t, h_S, and N_S, the coefficients a_i, b_i, c_i, d_i, the matrices A_{CN}, B_{CN}, and the obstacle $\mathbf{g}$ do not change with time. The product
$$A_{CN}\mathbf{g} \tag{201}$$
is also constant and can be precomputed once.

Use a warm start. A good initial guess is
$$\mathbf{x}^{(0)} = \max\left(\mathbf{U}^j - \mathbf{g}, 0\right), \tag{202}$$
because neighbouring time levels are close when h_t is small. This usually reduces the number of PSOR iterations.

Choose the relaxation parameter carefully. The parameter ω controls the degree of over-relaxation. The value $\omega = 1$ gives the projected Gauss–Seidel method. Values

$$1 < \omega < 2 \tag{203}$$

can accelerate convergence, but too large a value may slow convergence or cause oscillatory iterations.

Use an infinity-norm stopping rule. A practical stopping criterion is

$$\left\| \mathbf{x}^{(m+1)} - \mathbf{x}^{(m)} \right\|_\infty < \text{tol}. \tag{204}$$

The tolerance should be chosen consistently with the discretisation error; using an extremely small tolerance is not useful if the grid error dominates.

Check complementarity. A useful diagnostic is to compute

$$\mathbf{y} = A_{CN}\mathbf{x} - \widetilde{\mathbf{b}}^{j,j+1}. \tag{205}$$

At convergence, the numerical solution should approximately satisfy

$$\mathbf{x} \geq 0, \qquad \mathbf{y} \geq 0, \qquad x_i y_i \approx 0. \tag{206}$$

Extract the free boundary with a tolerance. The exercise region is detected by comparing $U_{i,j+1}$ with g_i. Because of round-off and iterative error, use a tolerance:

$$U_{i,j+1} - g_i \leq \varepsilon \tag{207}$$

for exercise, and

$$U_{i,j+1} - g_i > \varepsilon \tag{208}$$

for continuation. The free boundary can then be estimated by interpolation between the last exercise node and the first continuation node.

> **Optimisation summary**
>
> The main computational savings come from exploiting the tridiagonal structure, precomputing constant quantities, using the previous time level as a warm start, and projecting only the shifted variable $\mathbf{x} = \mathbf{U}^{j+1} - \mathbf{g}$.

Exercise and continuation regions

After computing $\mathbf{U}^{j+1}$, compare the option value with the obstacle.

The exercise region is

$$\boxed{\mathcal{E}^{j+1} = \{S_i : \ U_{i,j+1} = g_i\}.} \tag{209}$$

The continuation region is

$$\boxed{\mathcal{C}^{j+1} = \{S_i : \ U_{i,j+1} > g_i\}.} \tag{210}$$

The free boundary is the curve separating the exercise region from the continuation region. For an American put, early exercise is expected for sufficiently small values of S, while continuation is optimal for larger values of S.

Financial interpretation

For an American put, the option is exercised when the asset price is low enough that the immediate payoff $K-S$ is more valuable than waiting. The free boundary marks the critical asset price separating immediate exercise from continuation.

9 European put initial and boundary conditions

For a European put option,

$$V(S,T) = \max\{K-S, 0\}. \tag{211}$$

Therefore,

$$\boxed{U(S,0) = u_0(S) = \max\{K-S, 0\}.} \tag{212}$$

The boundary conditions are

$$\boxed{U(0,t) = u_a(t) = K\mathrm{e}^{-rt}, \qquad U(S^*,t) = u_b(t) = 0.} \tag{213}$$

On the grid,

$$U_{i,0} = \max\{K-S_i, 0\}, \qquad U_{0,j} = K\mathrm{e}^{-rt_j}, \qquad U_{N_S,j} = 0. \tag{214}$$

10 American put obstacle condition

For an American put option,

$$\boxed{U(S,t) \geq \max\{K-S, 0\}.} \tag{215}$$

Define

$$g(S) = \max\{K-S, 0\}, \qquad g_i = \max\{K-S_i, 0\}. \tag{216}$$

The continuous obstacle formulation can be written directly as

$$\boxed{\begin{aligned} &U(S,t) \geq g(S), \\ &\frac{\partial U}{\partial t} - \frac{\sigma^2}{2}S^2\frac{\partial^2 U}{\partial S^2} - rS\frac{\partial U}{\partial S} + rU \geq 0, \\ &(U(S,t) - g(S))\left(\frac{\partial U}{\partial t} - \frac{\sigma^2}{2}S^2\frac{\partial^2 U}{\partial S^2} - rS\frac{\partial U}{\partial S} + rU\right) = 0. \end{aligned}} \tag{217}$$

For the explicit method, one can project after the update:

$$\boxed{U_{i,j+1} \leftarrow \max\{U_{i,j+1}, g_i\}.} \tag{218}$$

For the implicit and Crank–Nicolson methods, the discrete problem is a linear complementarity problem:

$$\boxed{U_{i,j+1} \geq g_i, \qquad \text{finite-difference residual}_{i,j+1} \geq 0, \qquad \big(U_{i,j+1} - g_i\big)\text{finite-difference residual}_{i,j+1} = 0.} \tag{219}$$

11 Summary of schemes

Method	Time level used in spatial finite differences	Linear system?
Explicit	only j	No
Implicit	only $j+1$	Yes, tridiagonal
Crank–Nicolson	average of j and $j+1$	Yes, tridiagonal

Method	Time accuracy	Space accuracy
Explicit	first order, $O(h_t)$	second order, $O(h_S^2)$
Implicit	first order, $O(h_t)$	second order, $O(h_S^2)$
Crank–Nicolson	second order, $O(h_t^2)$	second order, $O(h_S^2)$